ABSTRACT

Complexity grows in multi-core setups and IoT uses, pushing more designs toward 3 -D Network-on-Chip systems for better internal data flow. Testing router functions matters a lot for stable performance; however old ways take too long and slow things down - especially when chips get bigger and stack in 3-D grids. To fix this, a smarter test approach using network coding is introduced here, tailored for 3 -D Torus NoCs. Instead of standard grid layouts, the Torus shape loops connections at edges, cutting how far signals must travel. Because of those loops, delays drop sharply across the chip. Routing shifts from basic Dimension-Order rules to a modulo-style system, which finds tighter routes while spreading traffic more evenly. Paths shrink. Bottlenecks fade. Efficiency rises without extra hardware. Instead of sending each packet alone, the new setup uses XOR coding inside routers to bundle several at once. This bundling cuts down how many routes are needed during checks, which shortens total test duration. Tests run through simulation show lower delays and better data flow rates than older mesh techniques. Hardware tests on FPGA boards confirm quicker error spotting alongside these gains. The design fits well within modern chip frameworks where power saving matters. Features like clock gating and power gating help reduce energy use across the network layers. Work happens directly on programmable chips to validate real-world function. Efficiency improvements appear without increasing complexity too much. Ideas apply mainly to systems built as single integrated units with shared communication backbones.